
Research Statement

I have long been interested in how AI/ML systems make connections across different types of information and apply what they have learned to new situations. A central question that motivates me is how machine learning models can generalize effectively beyond their training data and support reliable decision-making across domains. I am particularly drawn to the challenge of building systems that are not only accurate, but also reliable and capable of adapting when data is limited or contexts change. Through my academic studies and independent learning, I have developed an interest in how models represent knowledge and how these representations enable meaningful generalization and transfer. Generally, I am interested in machine learning-driven research at the intersection of data analysis, natural language processing, knowledge representation, and cognitively inspired approaches.

Motivated by this broad set of interests, I began with exploring a small aspect of the problem through my honours thesis, which is my primary research experience to date. To connect with my interest in how systems relate and transfer information, I examined how machine learning models handle analogical reasoning in natural language tasks. In this work, I designed and conducted experiments using modern language models, developed the analogical prompt dataset, and evaluated model behavior across different prompt types. Because the research question was defined by myself, the project required translating a vague problem into a clear experimental framework that allowed meaningful and fair comparison across models. Through this process, I gained insight into both the capabilities and limitations of current models, particularly with respect to generalization beyond training data and the ways knowledge may be represented in different forms.

Through this experience, I have developed a strong methodological foundation in machine learning and experimental design. I am proficient in Python and have experience with data preprocessing, statistical analysis, model evaluation, and visualization. I approach research as an iterative process, emphasizing clear problem formulation, careful experimentation, and thoughtful interpretation of results.

The Vector Institute represents an ideal environment for me to continue developing these interests. I am especially drawn to the opportunity to contribute meaningfully while learning from researchers working at the intersection of machine learning, applied systems, and societal impact, and to further develop my skills under mentorship in a collaborative research setting.

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